Central Limit Theorem

→ Given a sequence of identical i.i.d. rand. vars.

$$\underline{X}_1, \underline{X}_2, \dots, \underline{X}_n \text{ w/ } E(\underline{X}_i) = \mu, \ \mathrm{Var}(\underline{X}_i) = \sigma^2$$

⇒ The sample mean

$$\overline{X}_n = \frac{1}{n} \sum_{i=1}^{n} \underline{X}_i \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

↳ $SD = \frac{\sigma}{\sqrt{n}}$

Another way to state CLT:

⇒ The sum $S_n = \sum_{k=1}^{n} \underline{X}_k$

$$S_n \sim N(n\mu, n\sigma^2)$$